Benchmark Gifts: Four Measurable Workload Properties That Decided Agent-Mechanism Transfer in a Local-First Harness

Danilo Canivel
Independent / KAOS project
d.canivel@gmail.com

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Abstract

Like many teams building on LLM agents, we kept reading papers that promise self-improving agents—skill libraries, episodic memories, error-correction graphs—and kept facing the same question: if we build this into our harness, will it work on *our* workload? To stop guessing, we adopted a rule in KAOS, our open-source local-first agent harness: no new mechanism ships until it survives a kill test written and cryptographically locked before its code exists. A companion paper describes that machinery; this paper is its field report. Over the three months since v0.7 (May–July 2026) we ran thirteen mechanism candidates—eleven distilled from published papers—through the discipline, and a pattern emerged that the source papers never predicted: mechanisms died not because their ideas were wrong, but because our workload lacked something their benchmark had quietly provided for free. We name these hidden provisions *benchmark gifts* and identify four, each measurable on existing telemetry in an afternoon: a closed action vocabulary, dense outcome signals, lexically anchored task text, and per-task expert references. On our workload, two are absent (89.0% of agent trajectories with at least eight tool calls collapse to a single repeated action, in a workload dominated by one search-worker archetype; our outcome-signal tables hold zero organically produced rows, in part an adoption artifact we disclose), one is absent at the grain mechanisms need (exact expert-pair coverage 6.8%), and one is present (85.4% of task texts carry hard lexical anchors). Every measured outcome in our ledger tracked these axes—including a mechanism that scored a perfect 1.000 on an evaluation built from its own operating assumptions and still could not ship, because the property it depends on occurs in zero of twelve real trajectories. We offer this as a falsifiable hypothesis rather than a law, and a practical habit: before building a published mechanism, measure whether your workload supplies its gifts.

1 Introduction

In July 2026 we evaluated a mechanism that scored a perfect 1.000 on its evaluation workload, beat the incumbent baseline by 81 percentage points, passed every one of its four pre-registered kill gates—and was not shipped. The mechanism, a contrastive trajectory-diff localizer distilled from a published error-correction paper [Wang et al., 2026], had been probed under the falsifiable-evaluation discipline that governs KAOS development: kill gates written and cryptographically hash-locked before any feature code existed, a mandatory falsification self-test, and a binding verdict with no retune-and-rerun [Canivel, 2026]. The gate that should have protected against exactly this outcome—a requirement that real, organically produced agent trajectories exhibit the node-reuse structure the mechanism's premise consumes—*passed*, at a median of 10.0 against a floor of 1.30. A post-run instrument audit showed why: 97 of 109 organic agents run one tool in a loop, so their trajectories normalize to a *single repeated label*, satisfying a one-sided threshold vacuously. Among the twelve agents whose trajectories are diverse enough for a diff to operate on at all, median reuse was 1.000 and zero of twelve reached the floor. The mechanism's premise is false on the workload it was intended for; the verdict on file remains ACCEPT, the disposition is do-not-ship, and a regression test now fails if the mechanism ever enters the shipped package (§5.1).

We did not set out to write a paper about transfer; we set out to adopt good ideas from the literature, and the ledger of what actually happened to them became this paper. This paper is about the pattern that episode belongs to. Over the three months since v0.7 (May–July 2026) and thirteen mechanism candidates, the KAOS evaluation program has repeatedly watched mechanisms die not on their own logic but on the *shape of the workload*: an action-validation layer that could never be evaluated because the organic workload produced two qualifying incidents against a floor of two hundred [Xu et al., 2026]; a retrieval-side consolidation family rejected because lexical-only retrieval cannot simultaneously satisfy token-faithfulness and query-bridging [Canivel, 2026]; and now a trajectory-graph mechanism whose closed-vocabulary premise dissolves against real tool-argument entropy. Meanwhile the mechanisms that *did* survive measurement—BM25-based retrieval re-ranked by usage statistics—consume the one workload property our measurements show real tasks actually supply in abundance: hard lexical anchors.

We name the underlying variables *benchmark gifts*: properties that evaluation environments provide implicitly and gratis, that mechanism designs consume as unstated premises, and that real workloads may or may not supply. We propose four, all cheaply measurable: (i) a closed action vocabulary; (ii) dense outcome signals; (iii) lexically anchored task text; (iv) per-task expert references.

Contributions.

- *The four-axis workload-shape audit* (§3.3): closed action vocabulary (M1), outcome-signal density (M2), task-text lexical anchoring (M3), and expert-reference availability (M4), each with a deterministic, embedding-free measurement procedure, applied to a real harness's organic database (Table **??**).

- *A tiered evaluation ledger* (§4): thirteen mechanism candidates evaluated under one constraint set with explicit evidence tiers, extending the ten-candidate series of the companion paper [Canivel, 2026]; what is new here is the axis-level transfer account and the tier bookkeeping. Every *binding verdict* in the series was governed by hash-locked pre-registration; panel-tier judgments are labeled as judgments.

- *A probe lifecycle case study* (§5.1) in which a pre-registered gate passed vacuously, demonstrating a failure mode of pre-registration itself (one-sided predicates) and the instrument-audit practice that catches it.

- *A falsifiable synthesis* (§6): mechanism families mapped to the axes they consume, with the prediction—testable by any practitioner in an afternoon—that a mechanism fails to transfer when its consumed axis is absent.

This paper is the second of a pair, and the division of labor is deliberate. The companion paper [Canivel, 2026] describes the *machine*: the harness's plasticity architecture and the falsifiable-evaluation discipline that gates every change to it—the kill tests, the hash-locking, the binding verdicts. This paper is the machine's *field report*: what three months of those verdicts revealed about why published mechanisms fail to transfer, and which measurable workload properties decided each outcome. No new mechanism and no new discipline is introduced here; everything in this paper is measurement and pattern. Read the companion first if you want to trust the instrument; read this one first if you want to know what it caught.

Everything quantitative in this paper traces to a committed artifact in the public Kaos repository via a provenance ledger (Appendix A); claim labels (W1–W13) refer to its rows.

2 Related work

Do published agent mechanisms help in practice? The question has begun to receive direct negative evidence. SWE-Skills-Bench found 39 of 49 publicly available agent skills produced zero measurable gain on real software-engineering tasks, with three degrading performance by up to 10 points [Han et al., 2026]. CTIM-Rover found cross-task episodic memory failed to beat its baseline in any tested configuration, with retrieved items actively harming exploration [Lindenbauer et al., 2025]. HASP's own ablation showed unfiltered skill evolution costs 24 points [Liu et al., 2026]. These studies establish *that* transfer fails; our contribution is a measured account of *which workload properties* decide it, plus a discipline that catches the failure before shipping rather than after.

Harness engineering and the verification gap. The 2026 discourse converged on the position that the harness, not the model, is the active frontier [Hashimoto, 2026, Osmani, 2026, Zhou et al., 2026], with the sharpest critique being the absence of verification that harness changes help [Böckeler, 2026]. Recent systems work proposes evolution control planes that select among memory, skill, and weight interventions [Yan et al., 2026], but leaves the decision objective unoperationalized; Lee et al. [2026] optimize harnesses end-to-end against benchmark objectives, inheriting the benchmark-gift problem this paper measures. Evaluation methodology work has begun to quantify variance in agentic evals [Bjarnason et al., 2026]; we add the orthogonal axis: even a variance-correct benchmark result transfers only conditional on workload shape.

Pre-registration and falsification. Our discipline is Popperian pre-commitment applied to software mechanisms [Popper, 1959]: kill gates are written and hash-locked before feature code exists, a falsification self-test must prove the harness *can* reject the feature, and verdicts are binding. The companion paper describes the discipline and the plasticity architecture it governs [Canivel, 2026]—the machine; this paper treats that machine as a scientific instrument and reports what its output since v0.7 reveals about transfer—the field report. Classic self-improvement systems—Voyager's skill library [Wang et al., 2023], Reflexion's verbal reinforcement [Shinn et al., 2023]—are upstream of the mechanism families evaluated here.

3 Setting and method

3.1 The harness and its constraint set

Kaos is an open-source, local-first multi-agent harness whose entire runtime state resides in one SQLite file, with hard constraints relevant here: no embeddings, no parameter updates of any kind, no GPU requirement, no mandatory external services [Canivel, 2026]. The constraints matter methodologically: they force every adopted mechanism into a deterministic, auditable form, which is what makes the workload-shape measurements of §3.3 exact rather than sampled.

3.2 Evidence tiers

We report three tiers and label every claim: **Tier A**, pre-registered hash-locked probes with binding verdicts (two run to completion, two rejection-by-measurement studies); **Tier B**, deterministic mechanical measurements without a lock (the workload-shape audit; there is no mechanism under test, hence nothing to falsify); **Tier C**, structured panel evaluations of external papers (used only for motivation and candidate selection, never as evidence for a quantitative claim). The thirteen-candidate ledger (Table 1) spans all three tiers, and we consider the tier separation itself part of the method: a frequent failure of survey-style critique is treating read-the-paper judgments as if they were measurements.

3.3 The four axes

Each axis names a property that benchmark environments supply implicitly, together with the deterministic procedure we use to measure its presence in an organic workload (a live KAOS database: 453 agents, 1,639 tool calls, 171 task texts, accumulated by ordinary development and ARC-competition operation; composition disclosed in §7).

M1 — Closed action vocabulary. Text-game benchmarks admit exhaustive action normalization (e.g., `(take, apple, table)`); graph- and pattern-based mechanisms consume this by assuming repeated visits to identical normalized actions. *Measure:* normalize every tool call to a label—tool name plus a value-class signature over its arguments under fixed string rules (paths bucketed by top directory and extension, volatile keys dropped, long strings bucketed; frozen in the GDL probe lock, W5)—and compute per-agent *node reuse*: calls divided by distinct labels. *Result (W1):* 89.0% of agents with $\geq$8 calls are mono-label ($<$3 distinct labels; all 97 in this snapshot have exactly one)—largely a composition fact, since 86% of all calls come from a single search-worker archetype (§7); among the 12 label-diverse agents, median reuse is 1.000 (p75 = 1.120) and **0%** reach the 1.30 floor the GDL probe pre-registered. Organic trajectories are either one action repeated or all-unique chains; the middle regime that trajectory-graph mechanisms need was not observed (0/12; by the rule of three the one-sided 95% upper bound on its prevalence among diverse agents is ≈ 0.22).

M2 — Outcome-signal density. Benchmarks emit a reward every episode and often every step. *Measure:* count organically produced rows in the harness's outcome-carrying tables, and the tool-call error rate. *Result (W2):* the five plasticity tables (`episode_signals`, `skill_uses`, `failure_fingerprints`, `critical_steps`, `memory`) contain **zero** organically produced rows; 2 of 1,639 tool calls (0.12%) errored; the only outcome truth is episode-grain agent status (295 completed / 97 failed / 24 killed). The zeros are double-edged—they measure organic signal scarcity *and* incomplete adoption of the harness's own write paths during the development era—so the verdict we defend is scoped: the gift is absent *at sub-episode grain*, with the adoption reading disclosed rather than netted out (§7). This corroborates at population scale what a 2026 probe found at incident scale: an action-validation mechanism's binding evaluation returned VOID because the organic workload had produced n=2 qualifying incidents against a pre-registered floor of 200 (W6).

M3 — Task-text lexical anchoring. Benchmark task descriptions are templated, making task-similarity retrieval trivial for embeddings and hard for lexical methods—an inversion of the usual assumption. *Measure:* the fraction of task texts containing at least one hard lexical anchor (file path, snake_case/camelCase identifier, version number, error code, hex hash), by fixed regular expressions. *Result (W3):* **85.4%** (146/171) of organic task texts are anchored (paths in 144, identifiers in 127, versions in 124), with a median length of 656 tokens. Real operational tasks name their objects. This is the one gift the organic workload supplies *more* generously than benchmarks do—and it is

Candidate (source)	Tier	Outcome	Axis implicated
SAGE graph memory	C	reject (constraint conflict)	—
Synthesis-as-consolidation	A	**reject, measured** (W7)	M3†
Extractive consolidation	A	**do-not-build, measured** (W7)	M3†
AutoResearchClaw	C	orthogonal; 1 idea parked	—
HASP skill programs	C	do-not-build (external priors)	M2
Life-Harness action layer	A	**VOID #1, organic** n=2 < 200 (W6)	M2
UserHarness	C	park	—
Per-agent DB sharding	B	**reject by measurement** (W8)	—
MATM transactive memory	C	reject (embeddings); TRUF parked	—
ATG task graphs	C	park (6/10)	M1 (anticipated)
AReaL2.0 / ATDP	C	reject; 2 fragments distilled	M2
EMG $\rightarrow$ GDL probe	A	**accept-per-lock, do-not-ship** (W5)	M1
Retrieval re-ranking (ships)	B	**+10.0** / **+13.3 pp** (W9)	M3 (present)

Table 1: Thirteen mechanism candidates evaluated May–July 2026 (since v0.7, released 2026-04-15) under one constraint set. Tier A = pre-registered hash-locked probe; B = deterministic measurement; C = panel evaluation (motivation only). Bold = measured outcome. †The synthesis rejections established the *FTS vise*: under lexical-only retrieval a stored abstraction must be simultaneously token-faithful and query-bridgeable, which no tested variant satisfied (LLM arm 0.000 with the synthesized item present in top-K $\approx$50% of the time; extractive arm 0.000 despite 44/44 token faithfulness)—a structural boundary of the M3 axis rather than an absence of it.

consumed by exactly the mechanism family that has survived measurement in this program (W9).

M4 — Expert-reference availability. Correction-from-comparison mechanisms assume a per-task successful reference: EMG consumes one expert trajectory per training task from a curated dataset [Wang et al., 2026]. *Measure:* for each failed agent with a recorded task text, whether a completed agent exists with an identical task text (exact), or with token-Jaccard ≥ 0.5 (loose). *Result (W4):* exact coverage **6.8%** (5/73); loose coverage 68.5% (50/73). Loose pairs are common—but M1 makes them undiffable, so the gift is absent at the grain the mechanism family needs.

4 The ledger: thirteen candidates, three months

Table 1 summarizes the thirteen candidates evaluated since v0.7. Five were rejected, two closed do-not-build, three parked with documented re-entry conditions (panel records; bucketing: the mostly-orthogonal AutoResearchClaw row is counted as a park via its one parked idea, and MATM as a reject with its distilled TRUF fragment separately parked), one returned VOID (uninterpretable by lock design, a distinct and honest outcome), one was accepted-but-not-shipped (§5.1), and the retrieval mechanisms that ship in Kaos today entered before this series and were retroactively held to its measurement standards (W9). Zero mechanisms shipped on unverified promise. The right-hand column records the axis on which each evaluation turned, where one did.

Three regularities emerge, scoped to the measured rows. First, *neither Tier-A transfer probe died on the mechanism's internal logic*: both turned on workload gates, not capability gates—the capability claims (localization accuracy, validation lift) were either confirmed or never reached. (The sharding rejection is a premise-check on our own architecture, not a transfer failure; the FTS-vise rejections mark a structural boundary of the M3 axis rather than an absence of it.) Second, no measured outcome contradicted the axis account. Third, the single mechanism family with measured positive transfer consumes the single axis measured present.

5 Case studies

5.1 GDL: a perfect score, a vacuous gate, and a do-not-ship ACCEPT

The Graph-Diff Localizer probe (W5) is, to our knowledge, the first public end-to-end record of a pre-registered agent-mechanism evaluation in which the verdict and the shipping decision *diverged* under a no-retune constraint. The timeline, all commits public:

1. **Lock first.** Kill gates frozen and sha256-pinned at a commit containing no feature code: G1, accuracy ≥ 0.70 absolute and +10pp over the shipped single-trajectory localizer, $n \geq 40$; G2, a wrong-pair lesion isolating the pairing's causal role; G3, *organic* median node-reuse ≥ 1.30 (the reality gate); G4, lexical pairing precision ≥ 0.80. The workload was pre-registered as harness-generated (ground-truth divergence labels cannot exist organically), with the external-validity caveat written into the lock rather than discovered later.

2. **Falsification self-test.** Substituting the feature arm by the baseline (FULL := B1) emitted the required kill; substituting the lesion likewise. The self-test covered the capability gates (G1/G2); the workload gate G3 was force-passed inside the self-test harness—and G3 is precisely the gate later found defective, a scope limitation we return to below. A harness that cannot reject its feature is inadmissible; this one could, within that scope.

3. **Binding run.** FULL 1.000, B1 0.188 (below the seeded-random baseline B0's 0.208), wrong-pair lesion 0.000 (n=48); pairing precision 1.000; organic median reuse 10.000. All gates passed; verdict **ACCEPT** as computed. The B1 figure is itself a finding: the shipped heuristic localizer returns no answer on all 24 silent wrong-branch episodes because it keys on error steps, mechanically confirming the blind spot the mechanism targeted.

4. **Instrument audit.** The G3 median of 10.0 was implausibly high and was audited before the verdict was reported: 86% of organic tool calls come from search-worker agents that call one tool in a loop with a volatile-scrubbed argument, normalizing to a single label. A mono-label trajectory satisfies a one-sided reuse floor *vacuously*—and is exactly as undiffable as the all-unique chains the gate was written to catch. On the informative subpopulation (label diversity ≥ 3): median 1.000, **0/12** at the floor.

5. **Resolution.** Under the no-retune clause the gate was not edited and the run was not repeated; the verdict stands as computed. The *disposition* is do-not-ship: an ACCEPT whose only organic gate passed vacuously does not clear a shipping bar of "no mechanism ships on hope." Shipping requires a successor lock whose reuse gate excludes mono-label trajectories—a gate that on today's data would kill. A regression test enforces that no graph-diff module exists in the shipped package while the disposition stands (W13). The disposition followed rules since codified in the project's discipline document (commit `3d1b4d5`): every kill-gate statistic is audited after every binding run, passes and fails alike, and an audit may only hold or *downgrade* a disposition relative to the verdict, never upgrade it—discretion can cost the author's mechanism, never rescue it.

The methodological point: pre-registration constrains goalpost-moving but cannot guarantee a predicate measures what its author intended. The failure mode (one-sided thresholds satisfied by an unanticipated degenerate regime) is general, and the countermeasure—a mandatory post-run audit of every kill-gate statistic, with dispositions monotonically downgradeable but never upgradeable relative to the verdict (codified at commit `3d1b4d5`)—preserves both honesty and the no-retune property. We are not aware of an equivalent practice in published agent-mechanism evaluations.

5.2 VOID as data: the action-realization probe

The Life-Harness probe (W6) pre-registered an organic-only workload with a floor of 200 action-class incidents and explicitly forbade synthetic substitution. The organic count was 2. The binding verdict was VOID—not REJECT: the mechanism is unevaluated, not refuted—and the lock's synthetic ban is what made the outcome informative: it converted "we could not evaluate this" from a silent methodology hole into a measured statement about signal scarcity (M2) on real workloads. Population-scale corroboration arrived nine weeks later in the workload-shape audit: the outcome-signal tables that any such mechanism would learn from contain zero organic rows (W2).

5.3 The positive control

A transfer account that only explains failures explains nothing. The retrieval-re-ranking family—BM25 over verbatim text, re-ranked by Wilson-bound success statistics and recency [Wilson, 1927]—consumes M3 plus the one outcome signal M2 shows the workload does supply: lexically anchored text on both sides of the match, and success/failure at episode grain. It produced this program's only shipped mechanisms with measured gains (+10.0pp and +13.3pp top-1 over BM25 on small committed benchmarks—raw counts 8/10 vs. 9/10 and 11/15 vs. 13/15, one and two net flipped queries, directionally consistent but not statistically significant at these n; W9), and the GDL probe's pairing gate independently measured its premise at 1.000 precision on anchored task text. The axis account is consistent with both columns of the ledger—retrodictively; the circularity risk is confronted directly in §6.

6 Analysis: the audit, and what promotion gates miss

The workload-shape audit. The practical protocol this paper argues for costs an afternoon: before building a published mechanism into a production harness, measure the four axes on the target workload's existing telemetry (every measurement in §3.3 is a read-only query plus fixed string rules), then read the mechanism's evaluation section and list the gifts its benchmark supplied. A mechanism whose consumed axis is absent will not transfer, no matter how large its headline delta—GDL's was 81 points. The audit is falsifiable: it predicts, for instance, that EMG-family mechanisms transfer to workloads with repeated task families and low argument entropy (regression-test suites, scheduled runbooks), and fail on heterogeneous coding/ops traffic; and that skill libraries whose retrieval keys are anchored verbatim artifacts will outperform ones keyed on synthesized summaries on any M3-positive workload (the FTS-vise result, W7, is this prediction's boundary case).

The axes are retrodictive on this ledger. The four axes were formulated after watching these mechanisms die; on the ledger itself the account is retrodiction, and no statistical claim survives that circularity. Two pieces are nonetheless out-of-sample: the GDL reuse gate was pre-registered from a panel prior before the workload-shape audit existed, and the FTS-vise rejections predate the axis formulation entirely. The predictive content lives in the stated predictions above and in the audit protocol itself: any practitioner can measure the four axes on their own telemetry and check whether the next mechanism's fate tracks them. We publish the axes to be falsified, not because this ledger could confirm them.

Why promotion gates are not enough. The prevailing verification proposals for self-evolving agents—shadow evaluation, replay, canary, rollback [Yan et al., 2026]—are *promotion* gates: they ask whether a change helps on the traffic used to evaluate it. Every failure documented here would have sailed through them. GDL passes any promotion gate run on its constructed workload; the action-realization layer would have shipped with no evaluation at all (nothing in a promotion pipeline notices n=2); the synthesis mechanisms produced plausible artifacts that a canary would keep. The failures were caught by, respectively: a workload-shape gate plus instrument audit, a pre-registered sample-size floor with a synthetic ban, and kill gates written before the feature existed. The difference is Popperian: promotion gates accumulate confirmation; pre-registered probes purchase the right to ship by first proving the evaluation can refute.

Instrument risk is part of the method's honest accounting. Two of this program's most informative findings were defects in its own instruments: a 2026 self-audit found the verdict machinery could emit tautological agreement statistics and accept on empty gate sets [Canivel, 2026], and the GDL gate passed vacuously (§5.1). We report these not as embarrassments but as the discipline working: both were caught by self-audit—now mandated practice, codified at commit `3d1b4d5`, with the catches predating and motivating the codification—both produced mechanical guards, and neither required weakening a verdict already on file.

7 Limitations

One system, one workload. All Tier A/B evidence comes from a single harness and its operator's organic database. The workload is disclosed but idiosyncratic: 86% of tool calls come from ARC-competition search workers, which inflates the M1 mono-label rate; the label-diverse subpopulation behind the headline 0/12 figure has n=12. The axis *measurements* are exact; their *representativeness* is a single sample. The claim we defend is existential and historical—these axes are measurable and they tracked these thirteen outcomes—plus a falsifiable hypothesis for other workloads; not a universal law. The organic database is the operator's live working file and is not distributed; W1–W4 are verifiable against the committed snapshot values and regeneration scripts, but not re-runnable by third parties against our exact data—a standard telemetry limitation we flag rather than hide.

The axes are not proven exhaustive or independent. Four axes suffice to explain this ledger; other workloads may fail mechanisms on axes we have not needed (e.g., non-stationarity). M1 and M4 interact (loose expert pairs exist but are undiffable), so the axes are diagnostic dimensions, not an orthogonal basis.

Tier C is judgment, not measurement. Seven of thirteen ledger rows rest on structured panel evaluation of papers, not on probes. We use them only to characterize the candidate population and never cite a Tier-C number as evidence; still, candidate *selection* is a human choice and a source of bias in which mechanisms got probes at all.

M2's zeros are double-edged. Empty outcome tables measure both organic signal scarcity and the fact that this development-era database predates routine use of the plasticity write paths. Both readings are true; the second is why delayed-outcome backfill remains a live (parked) candidate rather than a refuted one.

Constructed-workload capability numbers do not claim external validity. GDL's 1.000/0.188/0.000 figures were pre-registered as harness-generated and carry that caveat by design; we cite them as evidence about the *baseline's blind spot* and the *probe lifecycle*, not as expected field accuracy.

8 Conclusion

Across thirteen mechanism candidates triaged under one constraint set—four of them through pre-registered probes or binding measurements—every measured transfer outcome was decided not by mechanism quality but by four measurable workload properties that benchmarks supply for free and real workloads may not. The practical consequence fits in one sentence: measure your workload's shape before you build—the queries are deterministic, the afternoon is cheap, and the alternative in our ledger was an 81-point benchmark win that would have shipped a mechanism whose premise occurs in zero of twelve real trajectories. Future work: replicating the axis measurements on other harnesses' telemetry (the protocol requires only tool-call logs and task texts); a successor GDL lock with a degeneracy-robust reuse gate; and probing the parked delayed-outcome backfill candidate, which targets the one axis (M2) that is absent today but is an artifact of write-path adoption rather than of the workload itself.

Acknowledgments and AI-use disclosure

Consistent with the program it describes, this paper was drafted by an agent operating inside Kaos, against the provenance ledger of Appendix A, and revised through a multi-level review panel (grounding, rigor, adversarial) before human sign-off; panel records are committed alongside the source. All quantitative content derives from the committed artifacts cited in the ledger. Responsibility for all claims rests with the human author.

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A Provenance ledger

Every quantitative claim in this paper carries a W-label resolving to a row of `paper2/PROVENANCE.md` in the Kaos repository (`https://github.com/canivel/kaos`), which maps the claim to a committed artifact (results file, lock file, verdict document, or commit hash) and an evidence tier. The workload-shape measurements (W1–W4) regenerate with `uv run python -m demo_workload_shape_bench.bench` against the operator's live database (not distributed; the committed `results.json` is the paper's frozen snapshot); the GDL probe (W5) with `uv run python -m demo_graphdiff_localizer_bench.run`; the test suite state (W13) with `uv run python -m pytest tests/`.